GridLink — Operating Cost Model

Monthly cost to keep the platform running at four customer sizes. Prepared September 30, 2026. Estimates, not quotes.

> **Definitions.** A "user" here is an **organization** — one buyer company or one supplier company — since that is the unit that transacts. Each organization has roughly 1–3 logins. The mix is assumed to be 60% buyers / 40% suppliers. All figures are monthly, in USD. Labor is "loaded" at base salary plus 15% for payroll tax and benefits.

1. Summary

Cost line	25 orgs	100 orgs	500 orgs	2,000 orgs
Hosting & infrastructure	$65	$140	$560	$3,100
AI tokens (Anthropic)	$40	$150	$750	$3,000
Transactional email (Resend)	$20	$20	$90	$300
Tooling & SaaS	$65	$120	$270	$600
Software subtotal	*$190*	*$430*	*$1,670*	*$7,000*
Labor (engineering + support)	$7,200	$22,600	$41,000	$97,000
Total operating cost / month	**$7,400**	**$23,000**	**$42,700**	**$104,000**
Cost per organization / month	$296	$230	$85	$52
Software-only cost per organization	$7.60	$4.30	$3.30	$3.50

Software is 2–7% of operating cost at every size. Labor is the rest. Infrastructure, AI and email together cost less per organization than a cup of coffee; the cost of running GridLink is the cost of the people who support and maintain it.

2. Hosting & infrastructure

Item	25	100	500	2,000	Basis
Vercel (app hosting)	$20	$40	$210	$800	Pro plan at $20/seat plus usage. Server actions and invoice PDF rendering drive function cost.
Supabase (database, auth, storage)	$25	$60	$175	$1,600	Pro plan $25 plus compute: Micro (included) → Small ($15) → Large ($110) → Team plan with 2XL compute and a read replica.
File storage & egress	$0	$5	$40	$150	~5 MB per supplier in compliance documents, RFP attachments, BOLs and invoice PDFs.
Domain, DNS, bandwidth buffer	$20	$35	$135	$550	Domain and DNS are a few dollars; the remainder is a buffer for CDN bandwidth and usage overage.
Hosting total	**$65**	**$140**	**$560**	**$3,100**	

Caveat. The current codebase has no pagination or caching, dashboards issue 20–30 database queries per load, and some page views trigger table-wide writes. Until those are fixed, expect to need the next Supabase compute tier roughly one size earlier than shown — about $50–100/month extra at each tier. After the fixes, the table above holds.

3. AI tokens

The platform calls Anthropic's `claude-opus-5` model, priced at $5 per million input tokens and $25 per million output tokens. Model reasoning ("thinking") is billed as output. Three features use it:

Feature	Effort setting	Approx. tokens per call	Cost per call	Calls per RFP	Cost per RFP
RFP draft (writes the RFP from a short description)	low	~2k in / ~1.5k out	$0.05	1	$0.05
Supplier ranking (scores candidate suppliers)	low	~6k in / ~3k out	$0.10	~3	$0.30
Bid brief (compares received bids for the buyer)	medium	~5k in / ~4k out	$0.13	~1.5	$0.20
Per RFP					**≈ $0.55**

At an assumed **4 RFPs per buyer per month**, that is **≈ $2.50 per buyer per month**. Suppliers use essentially no AI. The summary table applies this to the buyer share of each tier.

Supplier ranking is called ~3 times per RFP because it re-runs on every wizard interaction and is not cached. The bid brief *is* cached and only regenerates when bids change.

Reductions available

- **Cache the supplier ranking** per RFP draft. Drops the ranking line from $0.30 to $0.10 per RFP.

- **Move to `claude-opus-5-5`** ($4 / $20 per million tokens) — same model tier, 20% cheaper.
- Optionally run ranking on `claude-sonnet-5-5` ($2 / $10) and keep Opus for the bid brief buyers award on.

With the first two changes, AI cost falls to roughly **$1 per buyer per month**. Either way AI is small next to payroll — but it is the only line that scales per user with no ceiling, so per-user rate limits are a prerequisite.

4. Transactional email

Size	Emails / month	Resend plan	Cost
25 orgs	~4,000	Free tier / Pro	$20
100 orgs	~15,000	Pro (50k)	$20
500 orgs	~75,000	Scale (100k)	$90
2,000 orgs	~300,000	Scale with overage	$300

Volume is driven by invoice reminders, document-expiry reminders and RFP notifications — roughly 150 emails per organization per month once all notifications are wired up.

5. Tooling & SaaS

Item	25	100	500	2,000	Notes
Error tracking (Sentry)	$26	$26	$80	$250	Not currently in place; required before real customers.
Rate limiting (Upstash)	$10	$10	$30	$100	Protects public forms and AI endpoints.
Google Workspace	$28	$56	$112	$200	$14 per seat.
GitHub, Linear, Figma, 1Password	$1	$28	$48	$50	Free tiers at the smallest size; paid seats as the team grows.
Tooling total	**$65**	**$120**	**$270**	**$600**	

Not included — optional or conditional

- **SOC 2 compliance tooling** (Vanta or Drata): about **$1,000/month**, plus a $10–15k annual audit. Only needed when an enterprise buyer makes it a contract condition.
- **Fuel price index feed** (OPIS or Argus): roughly **$1,000–2,500/month**. Needed only if live rack-price benchmarking becomes a feature.

6. Labor

This is the cost to *keep the platform running and supported* — engineering and customer support. It does not include sales, marketing, the CEO, or any growth hiring.

Size	Team	Monthly (loaded)
25 orgs	CTO only ($75k base), covering engineering and support	$7,200
100 orgs	CTO + 1 senior engineer ($130k) + 0.5 customer-success ($60k)	$22,600
500 orgs	CTO + 2 engineers + 1.5 customer-success/support + on-call stipend	$41,000
2,000 orgs	CTO as engineering lead + 4 engineers + 1 SRE/DevOps + 4 customer-success/support	$97,000

Staffing rules of thumb used

- **1 support person per 150–250 B2B accounts.** Fuel buyers and dispatchers call; they do not file tickets.
- **1 engineer per 300–500 organizations** once the platform is stable, for maintenance, integrations and on-call.
- Salaries assume a mid-cost U.S. city and a remote team. San Francisco or New York rates would add roughly 25–35% to the labor line.

7. Metrics to watch

- **AI cost per buyer per month.** Log the `usage` figure returned by each call (it is currently discarded). If it drifts above $5, a cache has broken or an account is abusing the endpoint.
- **Supabase compute tier vs. customer count.** Needing an upgrade before ~300 organizations means the query fixes have not landed.
- **Support hours per organization.** Above 1 hour/month at 100 organizations indicates a product problem, not a staffing problem.

8. Assumptions

- Organization mix: 60% buyers, 40% suppliers. Buyers drive AI and RFP cost; suppliers drive document storage and email.
- Activity per buyer per month: ~4 RFPs, ~8 orders, ~8 invoices.
- Vendor pricing (Vercel, Supabase, Resend, Anthropic, Sentry) as published September 2026; subject to change.
- Labor loaded at base + 15%. No equity compensation, recruiting fees or one-time costs included.
- Excludes: sales and marketing staff, CEO compensation, legal, insurance, accounting, travel, office. Those appear in the fundraising budget, not in operating cost.
- All figures are planning estimates. Actuals should be reconciled against vendor invoices quarterly.